MMALS v1.1: From Functional-Memory Diagnostics to External Continual-Learning Evidence

RC2O-v2.2c SplitCIFAR10 Qualification, SplitCIFAR100 and CORe50 Diagnostics, and the v2.3 Context-Retention Rescue

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Abstract

Mycelium-Inspired Mutualistic Adaptive Learning Systems (MMALS) is a continual-learning research program organized around inferred context, functional routing, specialist hosts, replay and prototype memory, and explicit audit traces. This status article consolidates the path from MNIST-family functional-memory diagnostics to the latest external visual experiments. On SplitCIFAR10, the five-seed RC2O-v2.2c bridge reaches 0.91395 ± 0.00639 final average accuracy, versus 0.73860 ± 0.01559 for validation-selected replay, with paired gain $+0.17535$ and 95% confidence interval $[0.16093, 0.18977]$. On SplitCIFAR100, selected RC2O reaches 0.53556 ± 0.06207 versus 0.33254 ± 0.00742 for replay, but the representation/control ladder fails qualification. On a custom five-seed CORe50 object-level 10×5 stream, selected RC2O reaches 0.31490 ± 0.0345 versus approximately 0.2796 for replay, while severe early-task and old-context collapse prevents qualification. A two-seed v2.3 rescue ablation raises average accuracy from 0.2734 to 0.3171, minimum-task accuracy from 0.0050 to 0.0513, reduces mean zero-recall classes from 9.0 to 0.5, and breaks a one-host collapse; however, oldest-context top-1 remains zero, so the promotion gate fails. The evidence supports a robust external bridge on SplitCIFAR10, positive but bounded diagnostics on SplitCIFAR100 and CORe50, and a localized redesign of context-memory fusion. It does not establish end-to-end raw-pixel continual representation learning, state-of-the-art performance, positive backward transfer, or universal dataset generality.

1 Research question and contribution

The original MMALS question was not simply whether a modular neural network can achieve high classification accuracy. It was whether a learning medium can preserve useful transformations, infer latent context, route signals through reusable specialist hosts, and expose enough evidence to explain why a policy was selected. The research program therefore treats continual learning as a coupled problem of stability, plasticity, context inference, functional specialization, and governance.

The present contribution is a full status consolidation. It connects four evidence strata:

1. MNIST-family mechanism and geometry diagnostics;
2. a robust external bridge on SplitCIFAR10;
3. harder but representation-limited evidence on SplitCIFAR100;
4. a failure-driven CORe50 diagnosis followed by a targeted v2.3 rescue ablation.

The article deliberately separates a result from its maturity. A positive paired gain does not automatically imply robust retention, functional specialization, or benchmark qualification.

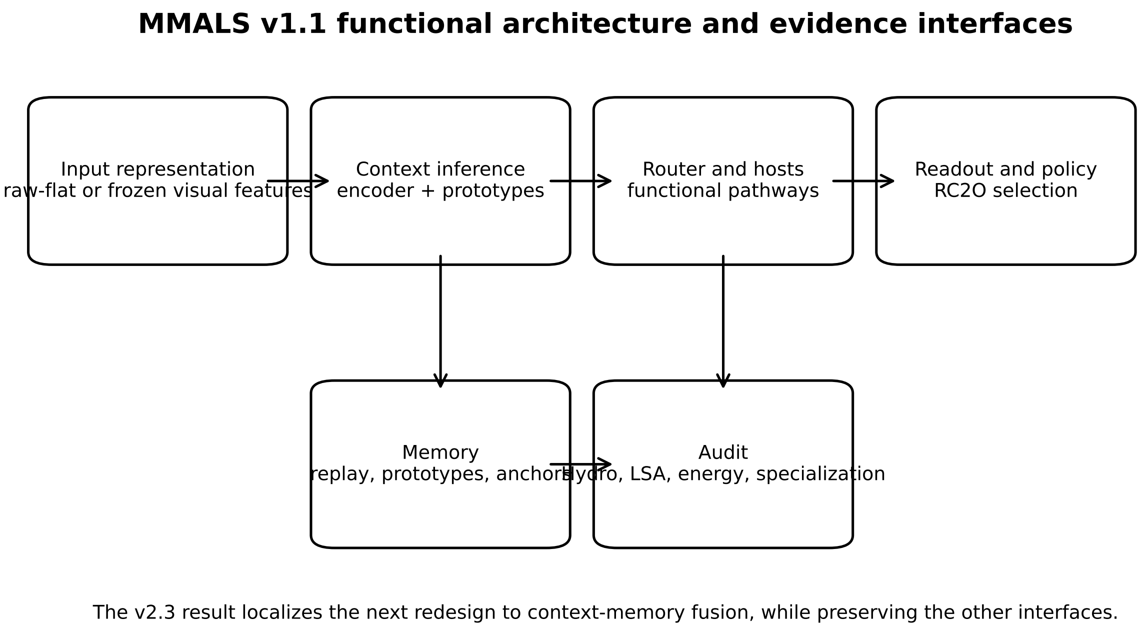


Figure 1: The current MMALS architecture separates input representation, context inference, functional routing, memory, audit, and deployment policy. The newest evidence localizes the next redesign to the context-memory interface rather than requiring replacement of the full backbone.

2 Architecture and learning object

Let $e = \phi(x)$ be an input representation. For MNIST-family studies, ϕ can be the historical flattened input path. For external visual studies, ϕ is a frozen supervised feature extractor. Each host H_k produces a transformed latent signal

$$z_k = H_k(e), \qquad k = 1, \dots, K. \tag{1}$$

A context posterior $q(c \mid e, m)$ and route vector $r(e, m, c)$ combine host outputs into a mediated latent state

$$z_f = \sum_{k=1}^{K} r_k(e, m, c)\, T_k(z_k), \qquad \sum_k r_k = 1, \tag{2}$$

followed by a global readout $p(y \mid z_f)$. The memory object is not route topology alone. Earlier route-function disentanglement experiments showed that two systems may share almost identical routes while computing different latent functions. Accordingly, MMALS audits route drift, mediated latent drift, output drift, prototype geometry, learning-signal allocation, and host ablation.

The RC2O controller is a validation-memory policy selector. It classifies the evidence regime and selects, locks, releases, falls back, or abstains among candidate policies. Final-test information is not used to trigger repair or choose the candidate.

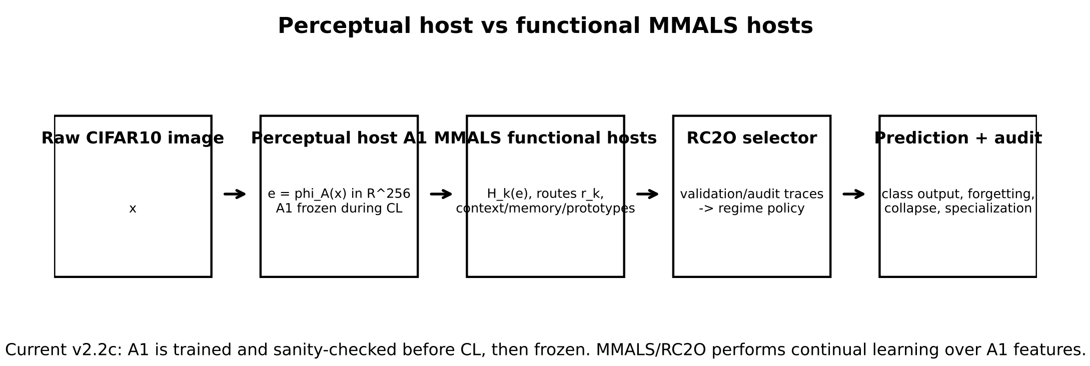


Figure 2: External visual bridge used in v2.2c: a supervised perceptual host is frozen before the sequential continual-learning experiment; the downstream MMALS functional hosts, memory and RC2O policy selector operate on the frozen features.

3 Evidence discipline

The program follows a progression of falsifiable gates rather than one aggregate score:

- **Representation sanity:** determine whether failures originate upstream of continual learning.
- **Baseline hardening:** select replay, EWC, LwF, PNN and MoE configurations using validation memory rather than arbitrary defaults.
- **Retention evidence:** report average accuracy, minimum-task accuracy, forgetting, oldest-task performance, minimum-class recall and zero-recall classes.
- **Context evidence:** report inferred-context accuracy, entropy, confusion, prototype separation and oracle-context gaps.
- **Specialization evidence:** report effective hosts, route dominance, host ablation and host probes.
- **Audit evidence:** retain Hydro, learning-signal alignment and energy probes as diagnostics unless explicitly activated for selection.

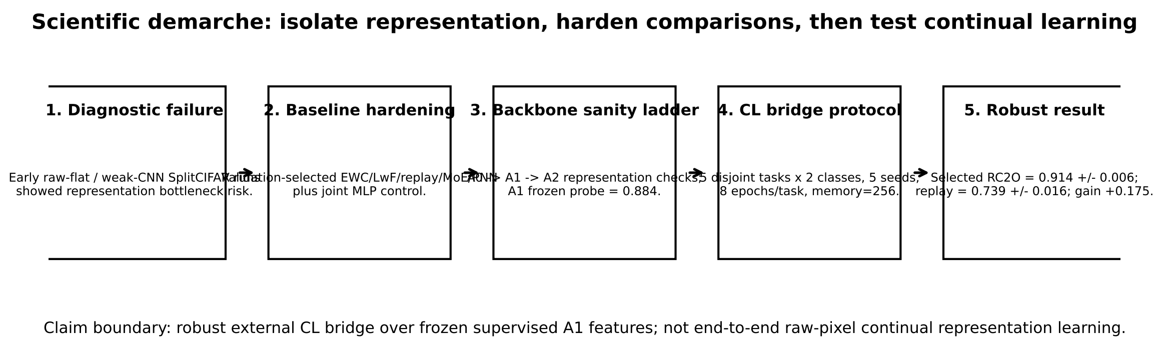


Figure 3: Scientific demarche used to move from representation debugging to bounded external continual-learning claims.

4 MNIST-family foundations

The MNIST-family branch established mechanisms before the external bridge. In v0.9, full functional memory reached approximately 0.9908 final average accuracy with average forgetting near 5.25×10^{-4}. The main lesson was diagnostic: route preservation alone could leave large latent-function drift, whereas latent-preserving variants tracked retention more faithfully. In v0.12, the geometry/co-evolution branch reached approximately 0.9889 final average accuracy and 0.0010 average forgetting, but the additional geometry regularizer did not clearly outperform the full functional-memory reference. This was retained as a microscope rather than promoted as a performance claim.

The later v1.1 selector line moved from explicit task identity toward inferred context. Its qualitative regression targets remain important:

- **MNIST:** clean context should permit a simple proto/global policy.
- **FashionMNIST:** ambiguous overlap should preserve the guarded context-gap family.
- **PermutedMNIST:** clean domain-shift evidence may release a true global guard.
- **RotatedMNIST:** weak-boundary evidence may justify a generic latent calibration candidate.

These behaviors show why a future universal context-memory module must support soft ambiguity and backward-compatible legacy operation rather than forcing hard context separation everywhere.

5 Consolidated evidence status

Table 1: Evidence strata and numerical status. Values across datasets are not direct difficulty-normalized rankings.

Dataset / branch	Protocol	Selected	Replay	Oracle	Status
MNIST-family v0.9	5-task functional-memory diagnostic	0.9908	–	–	Route-only memory was insufficient; functional latent preservation was supported diagnostically.
MNIST-family v0.12	geometry / co-evolution diagnostic	0.9889	–	–	Very low forgetting, but no clear accuracy advantage of the geometry regularizer.
SplitCIFAR10 v2.2c	5 tasks × 2 classes, 5 seeds	0.9140	0.7386	0.9235	Robust external-bridge qualification over frozen visual features.
SplitCIFAR100 v2.2c	10 tasks × 10 classes, 5 seeds	0.5356	0.3325	0.6551	Positive robust diagnostic; representation/control gate not passed.
CORe50 v2.2c	custom 10 tasks × 5 objects, 5 seeds	0.3149	0.2796	0.3631	Positive diagnostic; severe old-task and old-context collapse.
CORe50 v2.3	seeds 0 and 4, three rescue variants	0.3171	0.2734 ref.	–	Partial rescue: class and host collapse improved, oldest-context rescue failed.

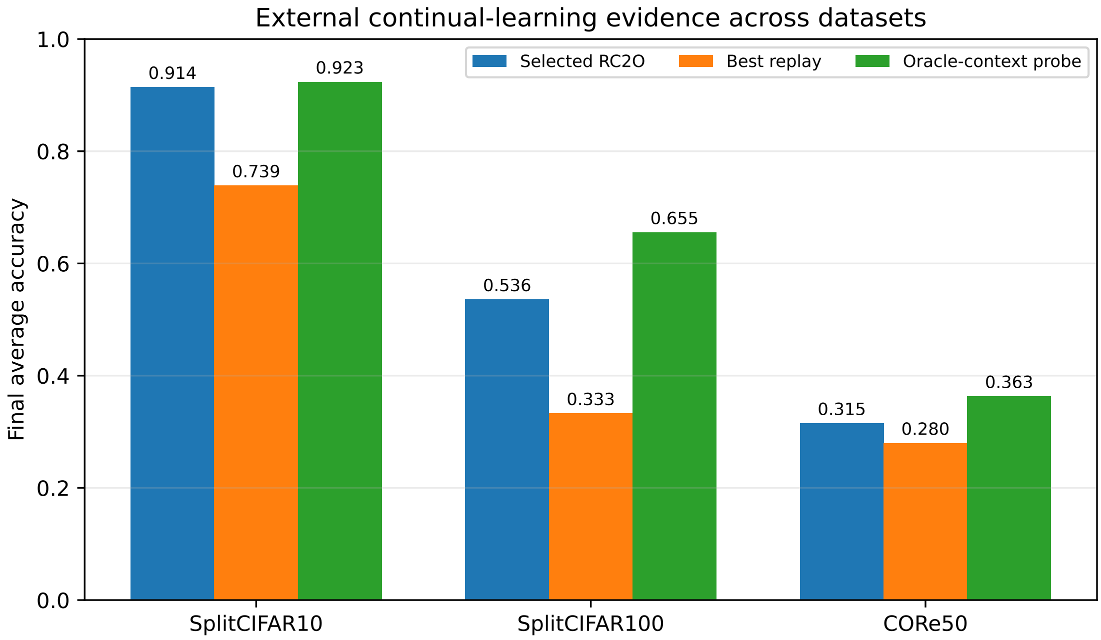


Figure 4: Selected RC2O, replay and oracle-context probe on the three external visual studies. Dataset difficulty and protocol differ; the figure is a status comparison, not a normalized leaderboard.

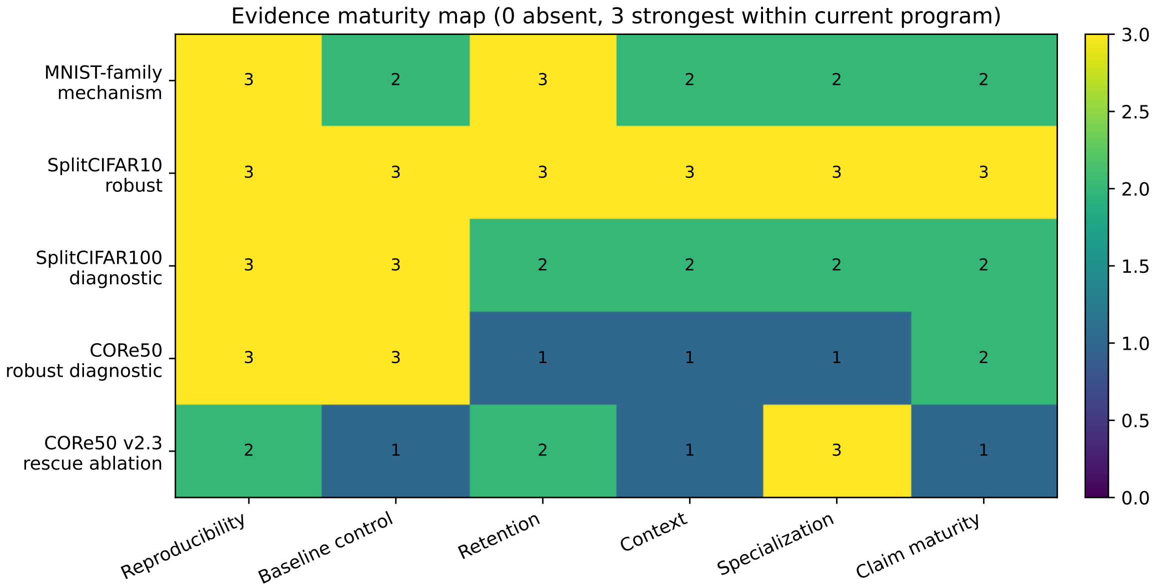


Figure 5: Evidence maturity map. The scoring is an internal program summary, not a standardized benchmark metric.

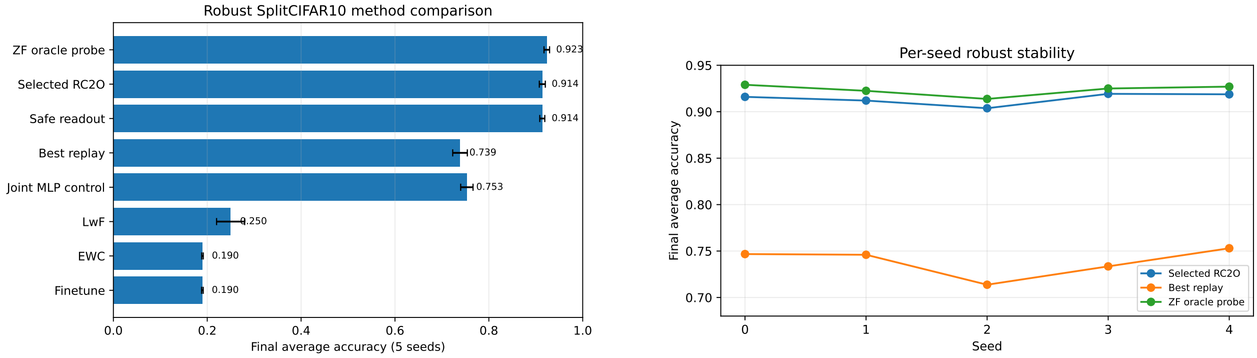
6 SplitCIFAR10: robust external bridge

The SplitCIFAR10 protocol uses five disjoint class-incremental tasks

$$(0,1) \to (2,3) \to (4,5) \to (6,7) \to (8,9). \tag{3}$$

A supervised A1 visual backbone maps images to a frozen 256-dimensional representation. The downstream experiment spans five seeds, eight epochs per task and validation-selected baseline sweeps.

Selected RC2O reaches 0.91395 ± 0.00639, compared with 0.73860 ± 0.01559 for the best replay baseline and 0.92345 for the oracle-context probe. The paired gain over replay is positive in every seed and averages $+0.17535$ with 95% confidence interval $[0.16093, 0.18977]$. Minimum-task accuracy is 0.8535, average forgetting is 0.02425, minimum-class recall is 0.8410, and no class has zero recall.



(a) Method comparison.

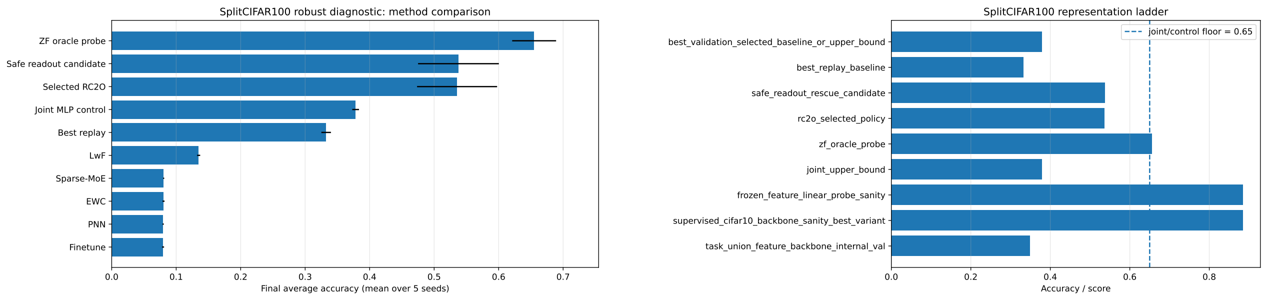
(b) Per-seed stability.

Figure 6: SplitCIFAR10 robust external-bridge result.

This is the strongest current non-MNIST result. The claim remains bounded: the visual representation is trained before the CL sequence and frozen. The experiment qualifies the downstream memory, routing, readout and selector bridge, not end-to-end continual feature acquisition.

7 SplitCIFAR100: positive but representation-limited diagnostic

The SplitCIFAR100 run uses ten tasks of ten classes. Selected RC2O reaches 0.53556 ± 0.06207, compared with 0.33254 ± 0.00742 for replay. The paired gain is $+0.20302$ with 95% confidence interval $[0.13431, 0.27173]$. The safe readout reaches 0.53770, while the oracle-context probe reaches 0.65506.



(a) Robust diagnostic comparison.

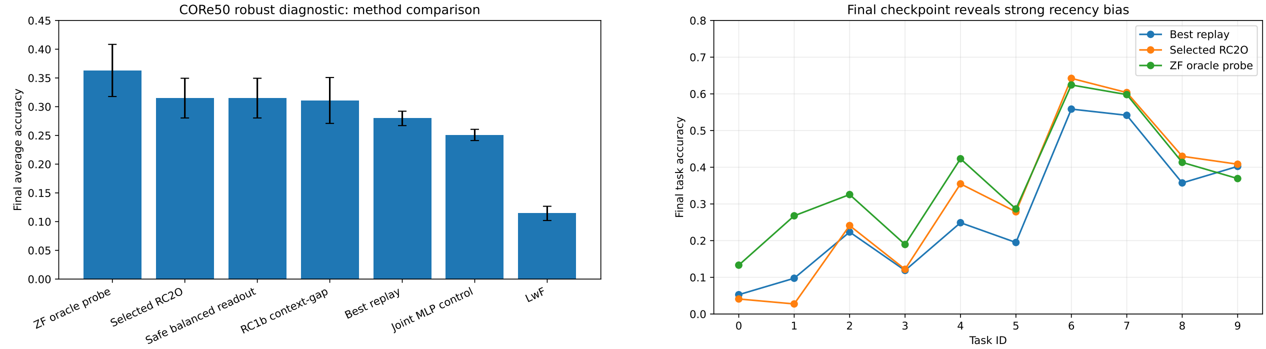
(b) Representation/control ladder.

Figure 7: SplitCIFAR100 result and qualification bottleneck.

The joint MLP control reaches only 0.37824, below the configured control floor. Therefore, the positive gain over replay is informative but cannot be separated cleanly from representation and readout limitations. The correct status is *positive robust diagnostic*, not qualification.

8 CORe50: transfer, failure and diagnosis

The CORe50 study uses a custom object-level stream of ten tasks with five object classes per task. The feature backbone is trained on train-only tensors, frozen, and followed by the same downstream MMALS/RC2O pipeline. Across five seeds, selected RC2O reaches 0.31490 ± 0.0345 and replay approximately 0.2796. The paired gain is $+0.03529$ with 95% confidence interval $[0.00292, 0.06766]$, positive in every seed.



(a) Five-seed method comparison.

(b) Final taskwise recency profile.

Figure 8: CORe50 robust diagnostic. Average accuracy improves, but earliest tasks nearly disappear.

The average conceals severe nonuniformity. Mean minimum-task accuracy is only 0.0146, compared with 0.0527 for replay. Early contexts are frequently assigned with near-zero entropy to the newest context. Yet the oracle-context probe reaches only 0.36305, about 0.048 above selected RC2O, so context error is not the only bottleneck. Latent-function preservation, replay balance, readout geometry and feature capacity also limit performance.

The specialization audit also reveals a seed with almost complete one-host collapse. Therefore, the run supports external transfer and a modest reproducible paired gain, but not strong retention, standard CORe50 benchmark comparability, or robust host specialization.

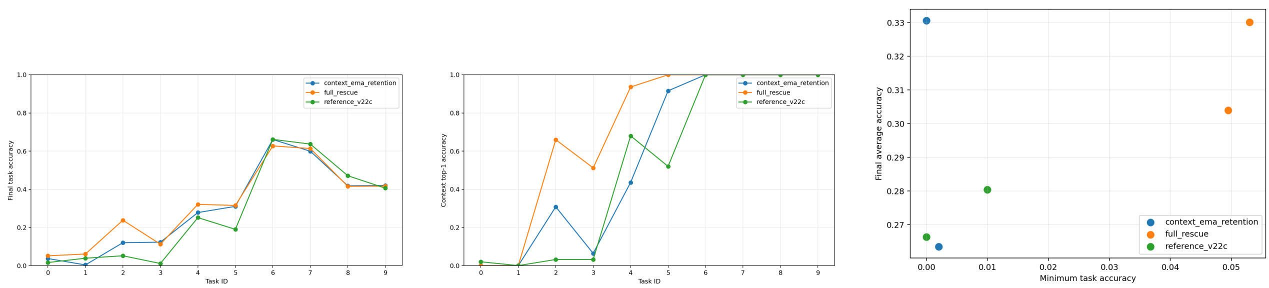
9 CORe50 v2.3: failure-driven rescue ablation

The v2.3 experiment does not repeat the unchanged five-seed run. It tests three variants on seed 0 and the route-collapse seed 4:

1. the v2.2c reference;
2. context EMA retention with old-task replay;
3. a full rescue combining stronger representation, ten hosts, host balance, replay, proxy repair and dynamic readout.

Table 2: Mean v2.3 rescue results over seeds 0 and 4.

Variant	Avg. acc.	Min task	Oldest 2	Context top-1	Zero recall	Forgetting
Reference v2.2c	0.2734	0.0050	0.0274	0.5284	9.0	0.2223
Context EMA retention	0.2970	0.0010	0.0203	0.5724	10.5	0.2101
Full rescue	**0.3171**	**0.0513**	**0.0560**	**0.7108**	**0.5**	**0.1499**



(a) Task retention.

(b) Context retention.

(c) Trade-off.

Figure 9: CORe50 v2.3 rescue diagnostics.

The full rescue raises final average accuracy by +0.0437, minimum-task accuracy by +0.04625, reduces average forgetting by about 0.0724, and reduces mean zero-recall classes by 8.5. Effective hosts increase from approximately 2.04 to 8.68, while maximum route dominance falls from 0.956 to 0.264. This is a substantive partial success.

However, the isolated context EMA variant reduces oldest-task retention and increases zero-recall classes. More importantly, oldest-two context top-1 remains zero under the full rescue. The promotion gate therefore correctly returns `iterate_before_robust`.

Table 3: Promotion-gate evidence.

Variant	Oldest-2 gain	Old-context gain	Zero-recall reduction	Decision
Context EMA retention	-0.0071	-0.0100	-1.5	Iterate
Full rescue	+0.0286	-0.0100	+8.5	Iterate before robust

10 What the newest run teaches

The v2.3 result isolates a mismatch between training and deployment. The retention loss directly trains a single-input context encoder, whereas the deployed posterior is dominated by a prototype-based branch. Consequently, the loss improves a component that contributes only weakly to the final decision, while the dominant prototype decision is not directly optimized by the retention objective.

This observation changes the next research question. The problem is no longer merely "increase EMA strength" or "add more replay." It is:

> How can the system preserve old context evidence while remaining plastic, train the posterior actually used at inference, and retain exact backward compatibility with prior results?

The proposed answer is a dataset-agnostic stable-plastic context-memory module with three operating modes: `legacy`, `stable_plastic`, and `adaptive`. The companion design note specifies the architecture, losses, regression gates and implementation contract. Importantly, the legacy path must reproduce earlier results without requiring users to recover historical notebooks.

11 Claim boundary

Table 4: Supported and unsupported external claims.

Evidence	Supported statement	Unsupported extrapolation
SplitCIFAR10 robust	RC2O substantially outperforms validation-selected replay with low forgetting and no class collapse over frozen A1 features.	End-to-end raw-pixel continual representation learning or CIFAR state of the art.
SplitCIFAR100 robust diagnostic	RC2O has a large paired advantage over replay.	Full qualification, because representation and joint-control floors remain weak.
CORe50 robust diagnostic	The external pipeline transfers and yields a modest positive paired gain in every seed.	Strong retention, standard CORe50 NC/NIC comparability, or proven host specialization.
CORe50 v2.3 rescue	Combined representation, replay, readout and host changes reduce zero-recall classes, forgetting and host collapse.	Attribution to context EMA or successful oldest-context retention.

No positive backward transfer claim is made. The CORe50 stream is custom and not equivalent to a standard NC or NIC benchmark. The current external experiments use frozen visual representations, so they do not validate sequential acquisition of unseen visual features. Several controls named "joint upper bound" in raw notebooks are reported here as *joint MLP controls*; because RC2O can exceed them, they are not theoretical upper bounds.

12 Program roadmap

The next sequence is deliberately conservative:

1. implement the stable-plastic context-memory module behind an exact legacy mode;
2. test mechanism variants on CORe50 seeds 0 and 4;
3. run one-seed regression smokes on MNIST, FashionMNIST, RotatedMNIST and PermutedMNIST;
4. protect the qualified SplitCIFAR10 result with strict non-regression gates;
5. revisit SplitCIFAR100 as a diagnostic with stronger representation and readout controls;
6. run a new five-seed CORe50 robust experiment only after context-specific gates pass.

13 Reproducibility package

The GitHub package contains this source and compiled PDF, selected figures, machine-readable evidence for SplitCIFAR10, SplitCIFAR100, CORe50 and v2.3, the key executed notebooks, claim-boundary notes, checksums, and build instructions. Large raw score dumps and full 100+ MB run archives are intentionally excluded from repository hygiene; the original run packages remain separate evidence artifacts.

14 Conclusion

The current evidence does not support the statement that MMALS has solved continual learning. It supports a more precise progression. Functional-memory diagnostics on the MNIST family established that route topology is not sufficient. SplitCIFAR10 provides robust external-bridge evidence over frozen visual features. SplitCIFAR100 and CORe50 show that the same architecture can transfer, but also expose representation, readout, context-retention and specialization limits. The v2.3 rescue substantially reduces class and host collapse yet fails its oldest-context target. That failure is scientifically productive: it identifies a localized, testable redesign while preserving the rest of the MMALS backbone and its audit discipline.

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